

SIMATS

ENGINEERING



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Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1-ASSESSMENT TOOL 2

COURSE CODE: Data Warehousing and Data Mining

COURSE NAME: CSA1601

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TOPIC 1: DECISION SUPPORT SYSTEMS (DSS)

INTRODUCTION TO DSS

- A Decision Support System (DSS) is a computer-based system that helps people make effective decisions.
- It collects and analyzes data to provide useful information.
- It supports complex and semi-structured decision-making.
- DSS is widely used in business, banking, and healthcare.

NEED FOR DSS

- DSS helps managers make accurate and timely decisions.
- It analyzes large amounts of data efficiently.
- It reduces uncertainty by comparing different alternatives.
- It supports planning, forecasting, and problem-solving.

COMPONENTS OF DSS

- Data Management: Stores and retrieves relevant data.
- Model Management: Uses mathematical and analytical models.
- User Interface: Allows users to interact with the system.
- These components work together to support decision-making.

TYPES OF DSS

- Data-driven DSS analyzes large amounts of business data.
- Model-driven DSS uses mathematical and statistical models.
- Knowledge-driven DSS provides recommendations using expert knowledge.
- Document-driven and communication-driven DSS support document analysis and collaboration.

DSS IN BANKING

- DSS helps banks evaluate loan applications.
- It supports credit risk assessment and financial forecasting.
- It helps identify suspicious transactions and fraud patterns.
- It assists banks in improving customer service and investment decisions.

DSS IN HEALTHCARE

- DSS helps doctors analyze patient information.
- It supports diagnosis and treatment planning.
- Hospitals use DSS to manage staff and medical resources.
- It helps healthcare organizations improve operational decisions.

Advantages and Limitations

- DSS improves decision quality and saves time.
- It supports data analysis and better planning.
- Implementation costs and technical complexity can be high.
- Incorrect data or unreliable models can produce poor recommendations

Topic 2: Operational Systems vs DSS

INTRODUCTION

- Operational systems manage an organization's daily business activities.
- DSS analyzes information to support managerial decisions.
- Operational systems focus on transaction processing.
- DSS focuses on analysis, forecasting, and planning.

OLTP CONCEPTS

- OLTP stands for Online Transaction Processing.
- It manages transactions such as payments, bookings, and purchases.
- It uses current and detailed operational data.
- Banking applications and online shopping systems are common examples.

OLAP CONCEPTS

- OLAP stands for Online Analytical Processing.
- It analyzes large amounts of historical business data.
- It supports complex queries and multidimensional analysis.
- Managers use OLAP to identify trends and compare performance.

KEY DIFFERENCES

- OLTP supports operational employees, while DSS supports decision-makers.
- OLTP mainly handles current transaction data, while DSS often analyzes integrated historical data.
- OLTP focuses on inserting, updating, and retrieving records.
- DSS focuses on analysis, comparison, and forecasting

PERFORMANCE COMPARISON

- OLTP is optimized for frequent transactions and short queries.
- DSS analytical queries may involve large datasets and complex calculations.
- Indexes and efficient database design improve system performance.
- OLTP prioritizes transaction speed, while DSS prioritizes useful analytical results.

USE-CASE SCENARIOS

- An ATM uses OLTP to process cash withdrawals.
- An online store uses OLTP to record customer purchases.
- A sales manager uses DSS to compare monthly revenue.
- A bank uses DSS to evaluate lending risks and business trends

INTEGRATION BETWEEN SYSTEMS

- Operational systems collect data from daily business transactions.
- ETL processes extract, transform, and load data into a warehouse.
- DSS and OLAP tools analyze the prepared data.
- Integration converts operational information into useful business insights.

TOPIC 3: ARCHITECTURE OF DATA WAREHOUSE

DATA WAREHOUSE DEFINITION

- A data warehouse is a centralized repository of integrated data.
- It collects information from multiple sources.
- It stores historical data for reporting and analysis.
- It supports business intelligence and decision-making.

THREE-TIER ARCHITECTURE

- The bottom tier contains data sources, ETL processes, and warehouse storage.
- The middle tier provides OLAP and analytical processing services.
- The top tier contains reporting, querying, and visualization tools.
- The three tiers organize data for efficient analysis and access.

Data Sources and Integration

- Data can come from databases, spreadsheets, and business applications.
- Different sources may contain different formats and structures.
- Data integration combines information into a consistent format.
- Integrated data improves the reliability of business reports.

ETL PROCESS

- Extract: Collects data from different source systems.
- Transform: Cleans, validates, and standardizes the data.
- Load: Stores the processed data in the warehouse.
- ETL improves data consistency and prepares information for analysis.

DATA MARTS

- A data mart is a smaller repository designed for a specific department.
- It can store information related to sales, finance, or marketing.
- Data marts provide focused access to relevant business information.
- They help departments perform analysis more efficiently.

METADATA

- Metadata means data that describes other data.
- It contains information about tables, columns, and data types.
- It records data sources and transformation rules.
- Metadata helps users understand and manage warehouse information

BENEFITS IN DECISION-MAKING

- A data warehouse provides integrated historical information.
- It improves reporting and analytical query performance.
- It helps identify trends and measure business performance.
- It supports forecasting, planning, and strategic decisions.

TOPIC 4: OLAP AND DATA CUBES

INTRODUCTION TO OLAP

- OLAP stands for Online Analytical Processing.
- It analyzes business data from multiple perspectives.
- It supports summarization, comparison, and trend analysis.
- It is widely used in business intelligence systems.

DATA CUBE STRUCTURE

- A data cube represents data using multiple dimensions.
- Dimensions may include Product, Region, and Time.
- Measures include sales revenue, profit, and quantity sold.
- Data cubes support multidimensional analysis of business information.

SLICE OPERATION

- Slice selects one specific value from a dimension.
- It reduces the number of dimensions being analyzed.
- Selecting sales for 2026 is an example of slicing.
- It helps users examine a particular part of the data.

DICE OPERATION

- Dice selects a subset of data using multiple conditions.
- It applies filters across two or more dimensions.
- Selecting laptop sales in Chennai during a specific quarter is an example.
- It helps users analyze specific business segments

ROLL-UPAND DRILL-DOWN

- Roll-up summarizes detailed data into higher-level information.
- Daily sales can be rolled up into monthly sales.
- Drill-down moves from summarized data to more detailed information.
- Yearly sales can be drilled down into quarterly or monthly sales

OLAPAPPLICATIONS

- Retail businesses use OLAP to analyze sales trends.
- Banks use OLAP to examine financial performance.
- Hospitals use OLAP to analyze patient and operational data.
- Manufacturers use OLAP to monitor production and inventory.

ADVANTAGES AND LIMITATIONS

- OLAP supports fast multidimensional data analysis.
- It helps users compare performance across different dimensions.
- Large datasets and complex queries may require significant resources.
- Data quality and suitable database design affect analytical accuracy.



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TOPIC 5: DIMENSIONAL DATA MODELING

INTRODUCTION TO DIMENSIONAL MODELING

- Dimensional modeling organizes data for reporting and analysis.
- It separates measurable business events from descriptive information.
- It commonly uses fact tables and dimension tables.
- It simplifies analytical queries in a data warehouse.

FACT TABLES

- A fact table stores measurable business events.
- It contains measures such as revenue, quantity, and profit.
- It includes keys that connect to dimension tables.
- Each row represents a defined level of detail called the grain

DIMENSION TABLES

- Dimension tables contain descriptive information about business entities.
- A Product dimension may contain product name and category.
- A Time dimension may contain day, month, quarter, and year.
- Dimensions allow users to filter and group facts for analysis

STAR SCHEMA

- A star schema contains a central fact table.
- Dimension tables connect directly to the fact table.
- Its structure is simple and easy to understand.
- It generally supports straightforward and efficient analytical queries

SNOWFLAKE SCHEMA

- A snowflake schema organizes dimensions into related sub-tables.
- It uses normalization to reduce duplicated descriptive data.
- It can require more joins than a star schema.
- It is useful when dimension hierarchies need separate representation.

SCHEMA DESIGN PROCESS

- Select the business process to be analyzed.
- Define the grain of the fact table.
- Identify the required facts and dimensions.
- Define keys, relationships, and data quality rules.

REAL-WORLD EXAMPLE

- A retail warehouse can analyze sales across multiple stores.
- The Sales fact table stores revenue and quantity sold.
- Product, Customer, Store, and Date tables provide descriptive details.
- Managers can compare sales by product, location, and month

TOPIC 6: QUERY PROCESSING AND BI TOOLS

INTRODUCTION TO QUERY PROCESSING

- Query processing converts a database query into executable operations.
- It identifies the data required to answer the query.
- The database system develops an execution plan.
- The system executes the plan and returns the results

STAGES OF QUERY PROCESSING

- Parsing checks the query syntax.
- Validation checks database objects and permissions.
- Optimization selects an efficient execution plan.
- Execution retrieves and processes the required data.

QUERY OPTIMIZATION

- Query optimization aims to reduce query execution cost.
- It compares possible execution plans.
- Indexes and suitable join strategies can improve performance.
- Efficient filtering and partitioning can reduce unnecessary data processing.

INTRODUCTION TO BI TOOLS

- A Business Intelligence tools transform data into useful information.
- They provide reports, charts, and interactive dashboards.
- They help users monitor key performance indicators.
- They support operational monitoring and strategic decision-making.

REPORTING AND VISUALIZATION

- Reports present structured information in tables and summaries.
- Bar charts compare values across different categories.
- Line charts display trends over time.
- Dashboards combine visualizations to support quick decisions.

OPEN-SOURCE BI TOOLS

- Apache Superset supports interactive dashboards and data exploration.
- Metabase provides accessible reporting and query-building features.
- KNIME supports visual data analysis through connected workflow nodes.
- These tools help users analyze data with different levels of technical expertise.

BENEFITS AND APPLICATIONS

- BI tools help organizations monitor sales and revenue.
- They support inventory tracking and customer analysis.
- Visual reports help users identify trends and unusual patterns.
- Reliable decisions require accurate data and correct interpretation.

TOPIC 7: OLAPAPPLICATION USING KNIME

INTRODUCTION TO KNIME

- KNIME is a visual platform for data analytics and integration.
- It allows users to build workflows using connected nodes.
- It supports data preparation, transformation, analysis, and visualization.
- It can be used to perform OLAP-style analysis of business data.

WORKFLOW COMPONENTS

- Nodes perform individual tasks in a workflow.
- Connections transfer data between nodes.
- Input nodes read data from files or databases.
- Processing and visualization nodes transform and present the results.

DATA IMPORT AND PREPARATION

- Import a sales dataset using a CSV or Excel Reader node.
- Check the columns and data types.
- Handle missing values and correct inconsistent records.
- Prepare clean data for grouping and analytical operations

ROLL-UPAND DRILL-DOWN

- Group sales by year to calculate annual revenue.
- Group sales by quarter to obtain more detailed summaries.
- Use monthly or daily grouping to examine sales in greater detail.
- These aggregation levels demonstrate roll-up and drill-down concepts.

SLICE AND DICE

- Use a Row Filter node to select data for a particular year.
- The filtered result demonstrates the slice operation.
- Apply multiple filters for region and product category to demonstrate dice.
- Compare the filtered results to understand specific business segments.

SAMPLE APPLICATION AND VISUALIZATION

- Use a retail sales dataset containing product, region, date, and revenue.
- Apply grouping and aggregation nodes to calculate total revenue.
- Create charts to compare regions and monthly sales trends.
- Use the results to identify high-performing products and locations.

REPORT GENERATION AND CONCLUSION

- Display analytical results using tables and charts.
- Create summaries of revenue, regions, and monthly trends.
- Use supported KNIME reporting features or export results as required.
- The workflow demonstrates how visual analytics can support business decisions.

THANK YOU